Fake News Classification: Style vs. Truth

DistilBERT vs. TF-IDF + Logistic Regression, with error analysis and adversarial probing

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Project Team

Name	Roll / ID No.
Soham Santra	2330196
Augnik Ganguly	2330079
Subhranil Ghosh	2304126
Arnav Singhal	2330293

1. Summary

This project asks whether text classifiers trained on labeled news can tell true stories from false ones, or whether they only recognize how a story is written. It compares two models on a balanced news corpus: a TF-IDF + Logistic Regression baseline and a fine-tuned DistilBERT transformer.

The main findings are:

- **Both models score very high in-distribution.** On a balanced 2,000-article test set, DistilBERT reaches 98.75% accuracy (ROC-AUC 0.9995) and Logistic Regression reaches 96.85% (ROC-AUC 0.9913).
- **DistilBERT makes 60% fewer errors** (25 vs. 63), with the biggest gain on false positives (4 vs. 33), meaning far less fake news is passed as real.
- **The high scores are largely a dataset artifact.** About 97% of real articles contain the token "reuters", versus about 1% of fake ones. Logistic Regression gives that token a coefficient of +9.00, the largest of any feature. The classifier mostly separates wire-service prose from hyper-partisan blog prose.
- **Style can override truth.** In five hand-written probes, an absurd claim in Reuters style was labeled REAL (99.9% confidence), and a verified event written as clickbait was labeled FAKE (99.7%).
- **The conclusion is sound, but some claims go further than the evidence.** Section 9 lists these, along with code and methodology issues worth fixing.

2. Project Contents

The project is a complete, reproducible experiment with a demo front end.

Component	Files	Role
Data	`data_loader.py`	Loads GonzaloA/fake_news, pools its splits, builds a balanced sample, makes stratified train/val/test splits

Component	Files	Role
Baseline	`train_baseline.py`	TF-IDF (10k features, 1-2 grams, sublinear TF) + Logistic Regression
Transformer	`train_distilbert.py`	Fine-tunes distilbert-base-uncased (66M parameters); keeps best checkpoint by validation accuracy
Analysis	`error_analysis.py`, `adversarial_probes.py`, `generate_plots.py`	False positive/negative breakdown, five style-vs-truth probes, charts
Orchestration	`run_pipeline.py`	Runs everything and writes consolidated_metrics.json
Scale-up	`train_welfake_big.py`, Kaggle notebook	Repeats the comparison on the 72,134-article WELFake dataset
Demo	`app.py, demo_predict.py`	Flask web lab and CLI for scoring any text with both models
Docs	README.md, REPORT.md, notebook	Narrative, earlier write-up, walkthrough notebook

Scope note: the uploaded files did not include the results/, models/ or templates/ folders. Metrics below are taken from the project's own REPORT.md and cross-checked for internal consistency against the code. The models were not re-run and the raw results JSON was not inspected.

3. Data and Method

Dataset. GonzaloA/fake_news (label 0 = fake, 1 = real). The loader pools the dataset's original train, validation and test splits, joins title and body as "title -- text", drops empty rows, and draws a class-balanced sample of 8,000 articles with seed 42.

Split	Size	Fake	Real
Train	5,000	2,500	2,500
Validation	1,000	500	500
Test	2,000	1,000	1,000

Baseline. TfidfVectorizer (10,000 features, unigrams and bigrams, English stop words removed, sublinear TF) feeding LogisticRegression (C=1.0, max_iter=1000). It trains in about 2 seconds on CPU.

DistilBERT. Sequence-classification head on distilbert-base-uncased; inputs truncated and padded to 256 tokens; AdamW (lr 2e-5, weight decay 0.01), batch size 16, linear schedule with 10% warmup, gradient clipping at 1.0, 2 epochs. Training took about 387 seconds on Apple MPS. The checkpoint with the best validation accuracy is evaluated once on the held-out test set.

Epoch	Train loss	Val loss	Val accuracy	Val AUC
1	0.1505	0.0545	98.20%	0.9982
2	0.0357	0.0678	98.30%	0.9991

4. Results

Metric	Logistic Regression	DistilBERT	Change
Accuracy	96.85%	98.75%	+1.90 pts
Macro F1	0.9685	0.9875	+0.019
ROC-AUC	0.9913	0.9995	+0.008

Metric	Logistic Regression	DistilBERT	Change
Fake precision / recall	96.99% / 96.70%	97.94% / 99.60%	
Real precision / recall	96.71% / 97.00%	99.59% / 97.90%	
Total errors (of 2,000)	63	25	-60.3%
False positives (fake called real)	33	4	-87.9%
False negatives (real called fake)	30	21	-30.0%
Training time	2.03 s (CPU)	387.39 s (MPS)	~190x

These numbers are internally consistent: the confusion-matrix counts reproduce every reported precision, recall and accuracy figure. As a rough uncertainty check, the 95% binomial confidence intervals on accuracy are about 96.1-97.6% for Logistic Regression and 98.3-99.2% for DistilBERT. They do not overlap, so DistilBERT's in-distribution advantage is likely real. A paired test such as McNemar's would be more precise.

Two caveats: the training-time comparison is not like-for-like (a CPU against an Apple GPU), and with a single seed and no repeated runs, the exact error counts could shift somewhat.

5. What the Models Learned

The Logistic Regression weights are the most direct evidence of what drives predictions.

Strongest cues for REAL: reuters (+9.00), said (+5.89), washington reuters (+3.94), weekday names such as wednesday (+2.64) and tuesday (+2.33), minister, spokesman, told reuters.

Strongest cues for FAKE: video (-4.96), image (-3.34), just (-3.11), featured image (-3.06), gop (-2.48), watch (-2.37), getty images, https, president trump, obama, hillary.

None of these is a truth signal. They are signs of source and format: a wire-service dateline and attribution style on one side, blog scraping artifacts (image captions, links, video embeds) and partisan vocabulary on the other. The corpus effectively contains two publishers, and the classifiers learned to tell them apart. That explains the high accuracy and why it should not be read as "detects fake news".

6. Error Analysis

DistilBERT made 25 errors on the test set: 4 false positives and 21 false negatives.

False positives (fake predicted real). Both examples reported in the project are fake articles written in a detached, institutional register. One was war-propaganda text in dry military prose (96.7% real); the other embedded an excerpt from a genuine New York Times piece (87.2% real). Journalistic phrasing outweighed an implausible premise.

False negatives (real predicted fake). None of the 21 false negatives contained "reuters", compared with 97% of real articles overall. The examples are opinion and cultural-commentary pieces in casual prose, scored 97-98% fake. Legitimate journalism that lacks wire conventions gets penalized, which is the same shortcut seen from the other side.

DistilBERT's remaining errors are mostly real articles flagged as fake (21 of 25). For a moderation use case, the model's residual risk on this corpus is over-blocking genuine reporting more than letting disinformation through.

7. Adversarial Probes: Style vs. Truth

Five short articles were written so that style and factual truth point in opposite directions, then scored by both models.

Probe	Truth	Style	Log. Regression	DistilBERT
P1 Moon core is dairy protein (IMF)	False	Reuters wire	REAL (89.9%)	REAL (99.9%)
P2 2013 Chelyabinsk meteor	True	All-caps clickbait	FAKE (85.2%)	FAKE (99.7%)
P3 28 GHz radiation damages tissue	False	Formal scientific	REAL (68.5%)	REAL (99.9%)
P4 Congress postpones winter	False (satire)	Political wire	REAL (90.7%)	REAL (99.9%)
P5 JWST mirrors aligned	True	Casual, unadorned	FAKE (83.9%)	FAKE (99.7%)

All five predictions follow style and none follows truth: both models fail all five probes. DistilBERT is also more confident than the baseline in every case, so greater accuracy in-distribution came with more overconfidence out-of-distribution.

8. Large-Scale Extension (WELFake)

train_welfake_big.py and the Kaggle notebook repeat the comparison on WELFake (72,134 articles from four merged sources), intended to dilute the single-source Reuters bias. Logistic Regression is trained on a full 80/20 split (TF-IDF with 25,000 features, C=2.0), while DistilBERT is fine-tuned on a subset: 10,000 articles for 1 epoch in the script, 20,000 for 2 epochs in the notebook.

No results from these runs appear in the uploaded files, and the README's mention of "40,000+ labeled articles" cannot be traced to a reported number. This is the most important gap: the project's central argument is that single-source bias drives the results, and the multi-source run is the natural test of that claim.

9. Critical Assessment

Strengths

- Clear, well-posed question with a clean baseline-vs-transformer design.
- Honest held-out evaluation: model selection uses validation data and the test set is touched once.
- Goes beyond accuracy to inspect coefficients, errors and adversarial cases, which is what makes the main conclusion credible.
- Complete, runnable code with a demo app and analysis scripts.

Limitations and claims to soften

1. **The probes are anecdotal, not proof.** Five hand-written items cannot establish a rate, and the notebook's wording ("empirical proof") is too strong. The probes also change several things at once: P2 contains "watch" (-2.37 in the baseline) and heavy capitalization; P1 and P4 carry the "(Reuters)" dateline, the baseline's top feature. A controlled design would present the same claim in both styles, or toggle one feature at a time.
2. **Mechanism claims about DistilBERT are speculation.** The earlier write-up says the dateline "dominates the transformer's attention heads", but no attention or attribution analysis was run. Only the Logistic Regression weights are directly interpretable.
3. **Possible duplicate leakage.** The loader pools all splits and resamples without deduplicating. News corpora of this kind often contain duplicate or near-duplicate articles, which can inflate test scores.

4. **Little evidence on truth beyond five probes.** The conclusion that classifiers cannot verify truth is well supported by the artifact analysis, but it is a claim about these models on this data; true-claim and false-claim behavior was only spot-checked.
5. **Single run, single split, single seed.** No variance estimate and no paired significance test, even though error_analysis.py already computes the discordant cases needed for McNemar's test.
6. **Balanced test set.** Real-world fake-news prevalence is not 50%, so reported precision would shift under realistic base rates.
7. **WELFake runs are not like-for-like.** Logistic Regression is scored on the full ~14k test split, while DistilBERT is scored on a 2,000- or 3,000-article subsample after training on a fraction of the data with different epochs. Also verify the dataset's label convention: WELFake's labels have been reported by users as inverted relative to its documentation, which would flip the meaning of "real" and "fake" in any probe interpretation.
8. **Source wording varies.** The README says real articles were "scraped directly from Reuters" (97.2%), while the report says 97.2% "contain the token reuters". These are different claims; the second is what was measured.
9. **The satire probe (P4) mixes categories.** Satire is neither fake news in the dataset's sense nor a verified falsehood, so its label is ambiguous.

Code and engineering notes

- app.py, demo_predict.py, run_pipeline.py and train_welfake_big.py hard-code a fallback virtualenv path under a personal home directory. Remove it or make it configurable.
- HF_HUB_OFFLINE=1 is set in the loaders, so the first run fails unless datasets and weights are already cached. Note this in the README.
- The WELFake notebook uses the deprecated torch.cuda.amp API and has no LR scheduler, unlike train_distilbert.py.
- The web app's explanatory warnings are keyword heuristics, not model outputs. This is fine for teaching but should be labeled as such in the UI.
- Inference pads every input to 256 tokens, which wastes compute on short inputs.

10. Recommendations

1. Report and compare the WELFake results on a single shared test set, and add cross-dataset evaluation (train on one source, test on another) to quantify how much accuracy collapses out-of-distribution.
2. Run a counterfactual experiment: strip datelines, source names and weekday markers from the real class and retrain, to measure how much of the accuracy depended on them.
3. Replace the five anecdotal probes with a larger, paired probe set (same claim, several styles) and report flip rates with confidence intervals.
4. Add a deduplication step and multiple seeds, and report McNemar's test between the two models.
5. Add an interpretability pass for DistilBERT (occlusion or gradient attribution) to support or drop the attention-based explanations.
6. If the aim is truth detection rather than style detection, move to a retrieval-based pipeline: claim extraction, evidence retrieval, natural-language inference, and source-reliability weighting. Treat a style classifier at most as a cheap pre-filter or a feature, never a verdict.

11. Conclusion

Fine-tuned DistilBERT clearly beats a TF-IDF + Logistic Regression baseline on the benchmark (98.75% vs. 96.85% accuracy, 60% fewer errors). But the baseline's weights, the error analysis and the probes all point the same way: both models recognize the look of Reuters-style reporting versus blog-style content, not whether a claim is true. High benchmark accuracy therefore says little about real-world fact-checking ability. The project's core message holds, and it would be considerably stronger with the multi-source results, cross-dataset tests, controlled probes and variance estimates described above.